

Ahmed Fathy

Software Engineer

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EDUCATION

- Cairo University, Faculty of Engineering** Giza, Egypt
B.Sc. in Computer Engineering; GPA: 3.7 2022 – 2027

EXPERIENCE

- Al Nada Scientific Office** | *Software Developer (Part-time)* Cairo, Egypt | June 2022 – 2025
 - ERP System:** Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
 - Corporate Web Solution:** Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

PROJECTS

- Tactic (Robotics & AI)** | *Embedded Systems* 2024
 - System Integration:** Developed browser-controlled robotics with AI vision and low-latency control
 - Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control.
 - Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
- Hankers (Social Platform)** | *Full-Stack Engineering* 2025
 - Architecture:** Built scalable social platform using Microservices principles, React.js, and TypeScript.
 - Frontend:** Implemented responsive UI design and asynchronous API integration via AJAX.
 - DevOps:** Established CI/CD pipelines and containerized services using Docker for consistent deployment.
- TCCD Official Website & Management System** | *Frontend Development — React, Typescript* 2026
 - Impact:** Developed a comprehensive student activity platform that successfully scaled to support over 20,000 active users.
 - Core Features:** Built a scalable event and judging system with slot management, waitlists, and role-based access.
 - Administration:** Developed an admin dashboard with analytics to manage events, users, and company profiles
- Image Restoration Pipeline** | *Deep Learning & Computer Vision, Python* 2024
 - Pipeline:** Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
 - Algorithms:** Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- Machine Fault Listener System** | *Machine Learning & Audio Processing* 2026
 - Pipeline & Architecture:** Designed and implemented an end-to-end machine learning pipeline to classify industrial audio signals into 6 distinct operational states (normal/abnormal).
 - Optimization:** Balanced model accuracy with inference speed under strict computational constraints (maximum 2B parameters), utilizing deep learning techniques for feature extraction and classification.
- Autonomous Wall-Following Robot** | *Embedded Systems & C/C++* 2026
 - System Architecture:** Designed a 5-state FSM to manage autonomous navigation, turning, and data flow.
 - Low-Level Firmware:** Programmed STM32 microcontroller at the register level to optimize hardware timer usage.
 - Control & Integration:** Combined proportional control, ultrasonic/gyro sensors, and WiFi Module for tracking.

STUDENT ACTIVITIES

- IEEE** | *UI/UX Designer - Frontend Team Leader* Cairo | Oct 2025 – Present
 - Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- NASA Space Apps Challenge** | *Participant – Project: "Orrery"* Cairo | Oct 2024
 - Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

TECHNICAL SKILLS

- Programming Languages:** C/C++, Python, JavaScript, TypeScript, SQL, PHP, Java, Assembly (x86/MIPS), OpenGL, GLSL
- Frontend & Full-Stack Web:** React.js, Next.js, Redux Toolkit, Tailwind CSS, Three.js, Node.js, Express.js, RESTful APIs, GraphQL, PostgreSQL, MongoDB, Laravel
- AI & Machine Learning:** PyTorch, TensorFlow, Keras, OpenCV, YOLO, Deep Learning, CNNs, Image Processing, NumPy, Pandas, Scikit-learn
- Robotics, IoT & Systems:** ESP32, Arduino, Unix/Linux Development, Multithreading, Memory Management, Network Socket Programming, WebSockets

- **DevOps & Core Concepts:** Docker, Git Version Control, CI/CD Pipelines, Object-Oriented Programming, Microservices, System Architecture, Algorithms